

Physical Peers

A Satellite-Based Measure of Firm Similarity

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Emerging Scholars in Accounting Conference

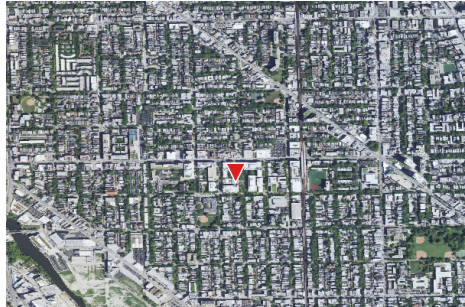


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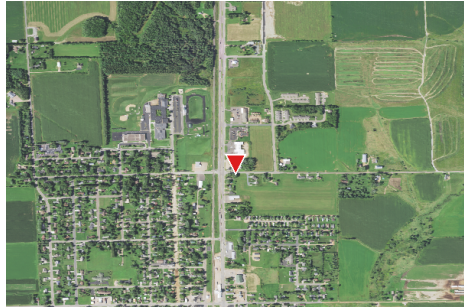
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Two Facilities, 1,157 km Apart



Apple (*electronics, NY*) Exxon Mobil (*petroleum, IL*) **1,157 km apart**

Two More, 1,185 km Apart



Sherwin-Williams (*chemicals, NY*) US Physical Therapy (*health care, WI*) **1,185 km apart**

Nothing in finance groups these firms together.

The Missing Relation

Finance

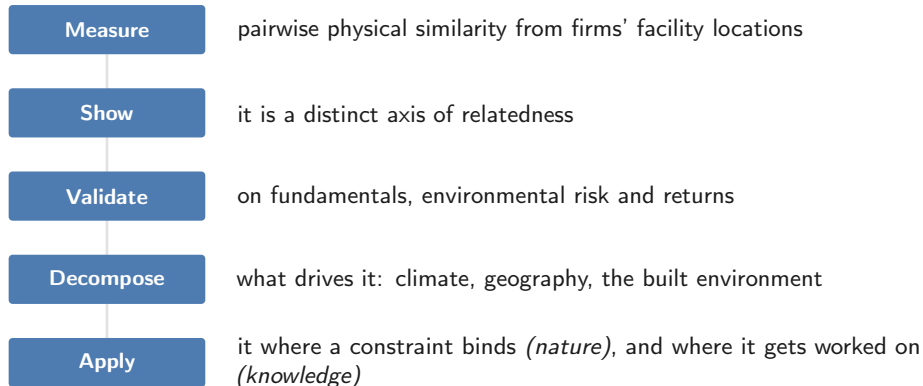
no market, no transaction, no analyst, no index puts these firms together

Satellite

the same configuration: land, water, built form, terrain, climate

This paper measures that relation, directly from satellite images of where firms operate.

This Paper



Contribution

What the literature assumes

Firms are related when they share a market, a transaction, or a labour pool
Hoberg & Phillips 2016; Lee et al. 2015

Physical risk is scored by what a firm depends on
Sautner et al. 2023; Garel et al. 2026

The environment enters only as a source of damage
Jaffe et al. 1993; Barrot & Sauvagnat 2016

What this paper shows

Show that two firms with matching footprints face the same constraints, without sharing any of those.

Separate the two: dependence is the firm's, condition is the place's. Show damage lands where they meet.

Show one configuration delivers both: where a constraint binds, damage lands together; where it is worked on, solutions travel.

Outline

- 01 Construction
- 02 Validation
- 03 Determinants
- 04 Applications
- 05 Takeaways

Outline

01 Construction

how a firm becomes a footprint, and how two footprints are scored

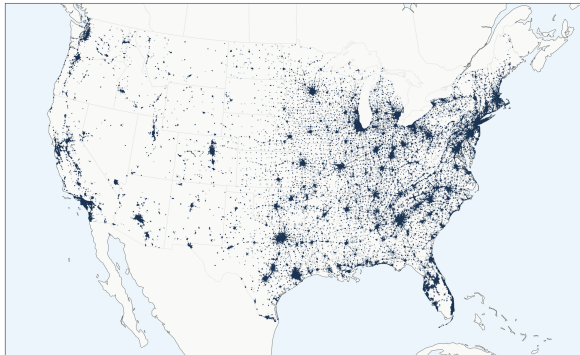
02 Validation

03 Determinants

04 Applications

05 Takeaways

Facility Locations



2024: 379,755 establishments of 1,634 firms.



129.1 M

Data Axle establishment records,
2017–2024

3.04 M

matched to a Compustat parent

2.71 M

with a satellite embedding

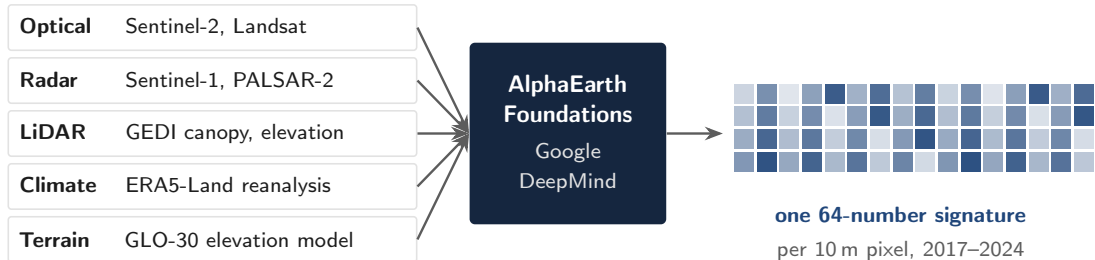
2,622 firms

12,552 firm-years with a footprint

2,482 firms

283,710 firm-pair-years in the
peer network

Satellite Embeddings

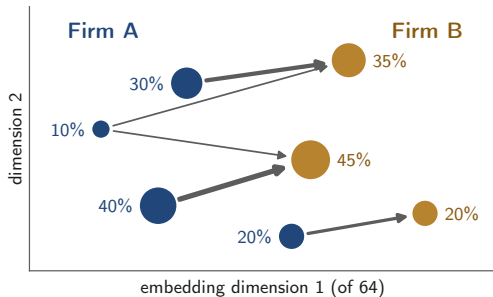


A vector records how built form, land use, terrain, water and climate combine at a site: its configuration, not any one condition alone.

A facility is the average vector within **1 km** of it: its surroundings, not the site itself.

Brown et al. (2025). Annual layers are public on Google Earth Engine.

Measuring Physical Similarity



A firm is its **footprint**: the places it occupies, weighted by employment.

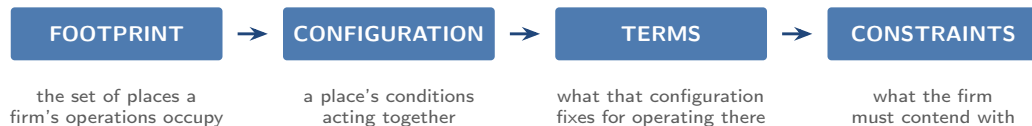
The distance between two firms is the **least mass** $\times$ **distance** that turns one footprint into the other.

Averages will not do: a firm split between city cores and farmland averages to somewhere suburban.

$$\text{EMD}_{ijt} = \min_{f_{mn} \geq 0} \sum_m \sum_n f_{mn} \|e_{imt} - e_{jnt}\| \quad \text{Similarity}_{ijt} = \frac{1}{1 + \text{EMD}_{ijt}} \in (0, 1]$$

subject to $\sum_n f_{mn} = w_{imt}$ and $\sum_m f_{mn} = w_{jnt}$; employment weights w sum to one within each firm-year.

What the Measure Captures



Two firms with matching footprints face the same constraints.

The measure is how far two footprints coincide. A **relation** between two firms, not a score on one.

Where a test needs a discrete set rather than a score, the annual threshold matches the density of SIC-3 (Hoberg and Phillips 2016). Firms above it are **physical peers**.

Outline

01 Construction

02 Validation

three tests of whether the firms the measure links actually behave alike

03 Determinants

04 Applications

05 Takeaways

Three Tests

- 1 Operating characteristics do peers combine inputs the same way?
- 2 Environmental risk does an outside assessor score them alike?
- 3 Stock returns does the market price them together?

Validation: Design

Outcome	closeness of a characteristic, $- x_i - x_j $: higher means more alike
Controls	TNIC-3 similarity, $ \Delta \text{size} $, $ \Delta \text{ROA} $, $ \Delta \text{leverage} $, same HQ state, log distance between facility centroids
Fixed effects	year, and 2-digit industry for <i>both</i> firms; SEs two-way clustered on both
Prediction	positive throughout, and it must survive restricting to different industries

The distance control is the one that matters: without it a positive coefficient could just be firms operating near one another.

Estimated on the peer network, where similarity has SD 0.042 against 0.064 across the candidate panel, a narrower range than the applications use.

Validation I: Operating Characteristics

	Tangibility	Cost intensity	Labour intensity ($\times 10^3$)
All physical-peer pairs	0.0091 *** (0.0017)	0.0103 *** (0.0039)	0.1768 *** (0.0473)
Pairs in different industries	0.0108 *** (0.0020)	0.0108 ** (0.0049)	0.2017 *** (0.0578)
Observations	281,816	281,816	280,733

→ Same magnitude across industries, and if anything larger. It is the configuration, not the sector.

Validation II: Environmental Risk

	Water	Biodiversity	Toxic emissions	Carbon
All physical-peer pairs	0.035 *** (0.013)	0.030 ** (0.013)	0.043 ** (0.019)	0.029 ** (0.015)
Pairs in different industries	0.026* (0.015)	0.029** (0.015)	0.037 (0.023)	0.030* (0.018)

→ MSCI builds these partly from asset locations, so this is convergent validity.

→ Across industries the point estimates hold but **precision falls**: toxic emissions drops outside conventional levels.

Validation III: Stock Returns

	Full network all peers	Cross-industry different SIC-2	Isolated different SIC-2 <i>and</i> HQ state
Physical-peer return	0.067 ^{***} (0.013)	0.062 ^{***} (0.009)	0.039 ^{***} (0.007)
Mean peer distance (log km)	0.000	0.000	0.000
Firm and industry $\times$ month FE	Yes	Yes	Yes
Observations	64,175	64,101	64,101

- Still priced when the peer shares neither industry nor HQ state.
- Peer distance is a precise zero, so this is not proximity.

Outline

01 Construction

02 Validation

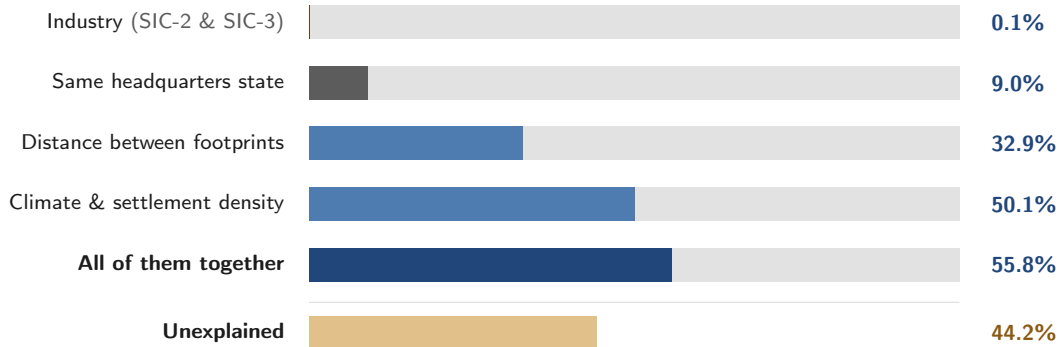
03 **Determinants**

which conditions it reflects, and how much of it none of them reach

04 Applications

05 Takeaways

Variance Decomposition



Each bar is how much of the measure that predictor explains. The rest is how the conditions **combine** at a site: the configuration.

What Peers Share

Mean absolute difference	Peers	Non-peers	Ratio
Distance between facility centroids (km)	451	1,341	3.0×
Freezing days (count)	12.0	47.6	4.0×
Annual precipitation (mm)	117	407	3.5×
Mean temperature (°C)	1.15	3.69	3.2×
Settlement density (log persons per km ²)	0.78	1.29	1.7×

→ Each condition is measured separately from the embedding.

→ Closeness on all of them at once is the configuration.

Reading the Scale

	Pairs	Mean similarity
Same firm, one year later	9,930	0.82
Same headquarters state	299,954	0.66
Same SIC-3, different HQ state	231,032	0.58
Different SIC-3 <i>and</i> HQ state	2,754,303	0.59

- The range is narrow, so every estimate is scaled per standard deviation.
- Across HQ states, sharing an industry does not make two firms more alike.

Outline

01 Construction

02 Validation

03 Determinants

04 Applications

what happens when a constraint binds, and what firms do to stop it binding

05 Takeaways

One Constraint, Two Sides

A configuration fixes the terms, and the terms decide what constrains a firm.

When it binds

What the firm brings is a **dependency**. What the place brings is the **condition** of the thing it depends on. Where both meet, damage lands together.

NATURE

When it is worked on

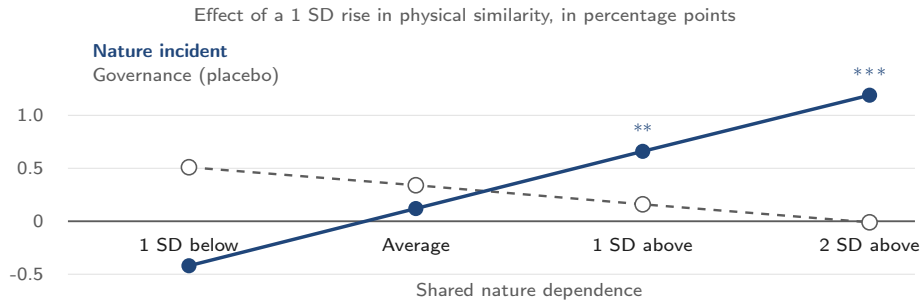
A constraint that gets worked on becomes a **solution**, and a solution is worth something to whoever else faces the same constraint.

KNOWLEDGE

Both are estimated on **all eligible pairs**, not only peers, with industry $\times$ year fixed effects for *each* firm and an internal placebo.

Application I: Correlated Nature Risk

- Outcome: **both** firms record a nature incident that year (13% of pairs; RepRisk, following Garel et al. 2026)
- Shared dependence: the *smaller* of the two firms' nature-dependence scores, so it is high only when both depend



- Nothing at or below average dependence. Neither ingredient works alone.
- At 2 SD the effect is 1.2 pp, about **9%** of the 13% base rate.

Application II: Knowledge Flow

- Outcome: either firm cites the other's patents (applicant-added, PatentsView)
- Sample removes the usual explanations: different SIC-3, headquarters **over 250 km** apart (2,224 km on average), and **never cited before**

	Applicant-added		Examiner-added (placebo)	
	All pairs any year	First links first ever	All pairs any year	First links first ever
Physical similarity	0.163 *** (0.033)	0.018 *** (0.005)	0.087*** (0.028)	0.005 (0.004)
Base rate (%)	0.642	0.068	0.610	0.058
Share of base rate	26%	27%	14%	n.s.

→ Tightening geography raises it: 0.196 beyond 1,000 km. Firm $\times$ year fixed effects on both sides: 0.100***.

Applicant vs. Examiner Citations

Applicant-added

Placed by the filing firm. Somebody there **found** the earlier work and drew on it.

26% of base rate

Examiner-added

Inserted by the patent office from its own prior-art search. Records that two inventions are **related**.

14% of base rate

Difference between the two gradients: **0.009** (0.003), $p = 0.004$.

**A shared configuration does not just set the same problem.
It moves the answer.**

Takeaways

- 1 Where production physically happens is **measurable**, and it relates firms that nothing else relates.
- 2 A **configuration** is how conditions combine at a site. Half of the measure is recognisable geography; no observable reaches the rest.
- 3 The firms it links behave alike in fundamentals, outside risk scores and prices, cross-industry throughout.
- 4 One configuration, two sides: where a constraint **binds**, damage lands together; where it is **worked on**, answers travel.

**Firms that share a configuration face the same constraints.
Damage lands on them together, and answers travel
between them.**

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Backup



Coverage by Year

Panel B: Coverage and network structure by year

Year	Firms	Pairs	Facilities per firm		Peers per firm		Same SIC-3 (%)
			Mean	Median	Mean	Median	
2017	1,171	22,750	217.3	10.0	38.9	20.0	13.3
2018	1,181	25,248	232.0	9.0	42.8	21.0	12.8
2019	1,310	32,296	230.9	9.0	49.3	25.5	14.0
2020	1,448	42,953	209.0	7.0	59.3	32.0	15.0
2021	1,581	48,676	199.8	6.0	61.6	33.0	15.3
2022	1,490	43,630	199.2	7.0	58.6	31.0	14.4
2023	1,433	37,865	212.6	8.0	52.8	28.0	13.9
2024	1,433	30,292	232.4	10.0	42.3	22.0	14.8

Distribution of the Measure

Panel C: Distribution of the measure

	N	Mean	SD	Min	p25	Median	p75	Max
<i>Candidate panel</i>								
Physical similarity	3,285,289	0.594	0.064	0.423	0.546	0.585	0.633	1.000
Earth Mover's Distance	3,285,289	0.703	0.175	0.000	0.579	0.710	0.833	1.365
<i>Physical-peer network</i>								
Physical similarity	283,710	0.728	0.042	0.679	0.698	0.715	0.745	1.000
Earth Mover's Distance	283,710	0.378	0.073	0.000	0.342	0.398	0.432	0.474

	Mean	SD	Correlation with	
			Log states	Log facilities
Typicality	0.584	0.022	+0.57	+0.55
Proximity	0.721	0.025	−0.56	−0.45

The Peer Threshold

Year	Firms	Candidate pairs	Kept pairs	Kept (%)	Similarity threshold
2017	1,171	319,702	22,750	7.1	0.6905
2018	1,181	326,945	25,248	7.7	0.6888
2019	1,310	370,462	32,296	8.7	0.6842
2020	1,448	436,125	42,953	9.8	0.6786
2021	1,581	487,465	48,676	10.0	0.6796
2022	1,490	483,628	43,630	9.0	0.6824
2023	1,433	466,904	37,865	8.1	0.6885
2024	1,433	394,058	30,292	7.7	0.6967
<i>All</i>	2,482	3,285,289	283,710	8.6	0.6862

Threshold matched each year to the density of SIC-3, following Hoberg and Phillips (2016).

Construction Choices

Choice	Why
1 km buffer	the configuration is around the site, not the roof
Employment weights	reflects where the people are
EMD, not a mean vector	city plus farmland averages to suburban
Facility-count decile matching	stops scale driving the distance
No residualisation	a standalone axis, not a residual

Uniform weights are reported in the paper, and are a *stronger* assumption.

One Firm, Two Comparisons

	Moderna (focal)	Philip Morris <i>Tobacco</i>	$ \Delta $	Elanco <i>Animal health</i>	$ \Delta $
<i>Physical similarity to Moderna</i>	–	0.79		0.54	
<i>Agglomeration & built environment</i>					
Built-up land (% of 1 km buffer)	72	76	5	19	53
Night-time lights ($\text{nW cm}^{-2} \text{ sr}^{-1}$)	77	77	0	17	60
<i>Land use</i>					
Cropland (% of 1 km buffer)	0	0	0	31	31
<i>Terrain</i>					
Elevation (m)	6	12	6	297	291
<i>Climate (control)</i>					
Mean temperature ($^{\circ}\text{C}$)	12	13	1	11	1

All three firms sit in roughly the same climate, so the grouping is not climate.

Variance Decomposition: Full Table

Model	Predictors	R^2	% of variation
<i>Standard groupings</i>			
(1)	Same HQ state	0.090	9.0
(2)	Same industry (SIC-2 & SIC-3)	0.001	0.1
<i>Facility-level geography</i>			
(3)	Distance between facility portfolios	0.329	32.9
(4)	+ same HQ state + same industry	0.353	35.3
<i>Climate and built environment</i>			
(5)	Temperature difference	0.201	20.1
(6)	Precipitation difference	0.275	27.5
(7)	Heat, flood, drought, freeze differences	0.395	39.5
(8)	Population-density difference	0.069	6.9
(9)	All climate + density	0.501	50.1
<i>All determinants</i>			
(10)	Climate + density + distance + HQ state + industry	0.558	55.8
<i>Unexplained by all determinants</i>			44.2

Geographic Distance: All Measures

Model	Geographic measure	% of variation explained		
		Linear	Logarithmic	20 bins
<i>Headquarters proxy</i>				
(1)	Same HQ state (indicator)	9.0	–	–
<i>Facility-level distance</i>				
(2)	HQ to HQ	19.8	22.1	23.6
(3)	Closest facility to closest facility	17.1	28.6	32.1
(4)	Footprint centroid to centroid	25.5	32.9	31.9
(5)	All three distances	–	43.7	49.0
(6)	+ same HQ state	–	43.8	–
<i>Unexplained by geographic proximity</i>			56.3	51.0

Overlap with Industry Classifications

	SIC2	SIC3	NAICS3	NAICS4	GICS4	GICS6
<i>Panel A. Correspondence by Year</i>						
2017	0.178	0.133	0.155	0.110	0.195	0.131
2018	0.176	0.128	0.150	0.103	0.191	0.121
2019	0.184	0.140	0.160	0.117	0.200	0.130
2020	0.195	0.150	0.171	0.129	0.205	0.132
2021	0.200	0.153	0.165	0.126	0.205	0.131
2022	0.191	0.144	0.153	0.119	0.202	0.130
2023	0.185	0.139	0.150	0.115	0.190	0.124
2024	0.196	0.148	0.157	0.118	0.204	0.135
Average	0.188	0.142	0.158	0.117	0.199	0.129
<i>Panel B. Correspondence by SIC Groupings</i>						
Agri, forestry, fishing	0.000	0.000	0.004	0.001	0.066	0.062
Mining	0.351	0.281	0.282	0.280	0.413	0.318
Construction	0.071	0.070	0.071	0.070	0.154	0.081
Manufacturing	0.200	0.157	0.216	0.155	0.220	0.136
Wholesale trade	0.060	0.022	0.056	0.022	0.130	0.062
Retail trade	0.070	0.047	0.060	0.048	0.185	0.148
Services	0.213	0.150	0.072	0.053	0.137	0.091
Public administration	0.004	0.004	0.004	0.004	0.156	0.004

Network Persistence

	Firm-years	Mean	p25	Median	p75
<i>Panel A: Overlap of a firm's peer set across horizons</i>					
One year later	8,407	0.45	0.29	0.46	0.60
2 years later	6,569	0.33	0.17	0.31	0.44
3 years later	4,983	0.27	0.12	0.23	0.36
4 years later	3,611	0.24	0.09	0.18	0.33
5 years later	2,438	0.21	0.06	0.15	0.29
6 years later	1,477	0.18	0.04	0.12	0.27
7 years later	681	0.16	0.03	0.10	0.25
<i>Panel B: One-year overlap, by change in the facility portfolio</i>					
Facility count changed by 10% or less	5,722	0.51	0.37	0.50	0.65
Facility count changed by more than 10%	2,685	0.33	0.15	0.32	0.50
<i>Panel C: Similarity of a firm's own facility footprint, t against $t + 1$</i>					
Same firm, one year later	9,930	0.82	0.79	0.83	0.86

Validation I: Full Table

<i>Panel A: All physical-peer pairs</i>			
	Tangibility (1)	Cost Intensity (2)	Labor Intensity ($\times 10^3$) (3)
Physical similarity	0.0091*** (0.0017)	0.0103*** (0.0039)	0.1768*** (0.0473)
Product-market similarity (TNIC)	0.0192*** (0.0020)	0.0161*** (0.0027)	0.3918*** (0.0576)
Δ Log assets	-0.0075** (0.0034)	-0.0170*** (0.0044)	-0.2837*** (0.0706)
Δ ROA	0.0138** (0.0070)	-0.1609*** (0.0348)	0.4566** (0.1885)
Δ Leverage	-0.0082 (0.0189)	-0.0819*** (0.0297)	1.5463*** (0.5964)
Same headquarters state	0.0202** (0.0091)	0.0389** (0.0166)	0.8392*** (0.1783)
Log distance between facility centroids	0.0027* (0.0015)	0.0026 (0.0041)	0.1046*** (0.0400)
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Observations	281,816	281,816	280,733
Adjusted R^2	0.243	0.259	0.353

Validation II: Full Table

<i>Panel A: All physical-peer pairs</i>				
	Water (1)	Biodiversity (2)	Toxic Emissions (3)	Carbon (4)
Physical similarity	0.0352*** (0.0126)	0.0296** (0.0125)	0.0428** (0.0193)	0.0293** (0.0146)
Product-market similarity (TNIC)	0.2107*** (0.0124)	0.1485*** (0.0130)	0.2245*** (0.0146)	0.1565*** (0.0113)
Δ Log assets	-0.0050 (0.0104)	-0.0158 (0.0097)	-0.0013 (0.0165)	-0.0066 (0.0122)
Δ ROA	0.1735** (0.0739)	0.1423 (0.0991)	0.3231*** (0.1127)	0.2444*** (0.0855)
Δ Leverage	-0.0226 (0.0359)	-0.0210 (0.0451)	-0.0246 (0.0465)	-0.0403 (0.0314)
Same headquarters state	0.0918** (0.0382)	0.1204*** (0.0423)	0.1846*** (0.0539)	0.1812*** (0.0459)
Log distance between facility centroids	-0.0041 (0.0096)	-0.0166* (0.0093)	-0.0123 (0.0119)	-0.0146 (0.0124)
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Observations	166,494	163,895	166,102	166,483
Adjusted R^2	0.248	0.283	0.194	0.382

Validation III: Full Table

	Full network (1)	Full network (2)	Cross-industry (3)	Isolated (4)
Physical peer return (full network)	0.067*** (0.013)	0.091*** (0.013)		
Physical peer return (cross-industry)			0.062*** (0.009)	
Physical peer return (isolated)				0.039*** (0.007)
Mean peer distance, full network (log km)	0.000 (0.002)	-0.001 (0.001)		
Mean peer distance, cross-industry (log km)			0.000 (0.002)	
Mean peer distance, isolated (log km)				0.000 (0.002)
Industry benchmark return (VW)		0.422*** (0.030)		
State benchmark return (VW)		0.121*** (0.026)	0.097*** (0.025)	0.104*** (0.025)
Firm fixed effects	Yes	No	Yes	Yes
Time fixed effects	No	Yes	No	No
Industry $\times$ time fixed effects	Yes	No	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Observations	64,175	64,286	64,101	64,101
Adjusted R^2	0.248	0.181	0.248	0.248

Placebo: Shuffled Measure

	Physical	Shuffled	
	similarity (1)	Stratified (2)	Unrestricted (3)
Tangibility	0.0091*** (0.0017)	0.0041*** (0.0009)	-0.0003 (0.0003)
Cost intensity	0.0103*** (0.0039)	0.0025 (0.0023)	-0.0002 (0.0008)
Labor intensity ($\times 10^3$)	0.1768*** (0.0473)	0.0840*** (0.0253)	-0.0082 (0.0078)
MSCI water stress	0.0352*** (0.0126)	0.0297*** (0.0078)	-0.0051* (0.0030)
MSCI biodiversity	0.0296** (0.0125)	-0.0014 (0.0069)	-0.0004 (0.0023)
MSCI toxic emissions	0.0428** (0.0193)	0.0259** (0.0104)	-0.0038 (0.0031)
MSCI carbon	0.0293** (0.0146)	0.0116 (0.0092)	0.0028 (0.0025)
Outcomes significant at 5%	7 of 7	4 of 7	0 of 7
Correlation with the physical measure	—	0.228	0.003

Column 2 keeps location and size and correlates 0.23 with the real measure, so it is **not a null**. Column 3 is.

Application I: Full Table

<i>Panel A: Interaction estimates</i>			
	Nature incident		Placebo
	All pairs (1)	Cross-industry (2)	Governance (3)
Physical similarity	0.0012 (0.0025)	0.0000 (0.0025)	0.0034 (0.0027)
Shared nature dependence	0.0013 (0.0036)	0.0008 (0.0036)	−0.0037 (0.0032)
Physical similarity × Shared dependence	0.0054*** (0.0016)	0.0054*** (0.0016)	−0.0017 (0.0017)
Own incident-rate guard	Yes	Yes	Yes
Firm-difference controls	Yes	Yes	Yes
Geographic distance control	Yes	Yes	Yes
Cross-industry pairs only	No	Yes	No
Industry × year FE	Yes	Yes	Yes
Observations	321,680	314,916	321,680
Adjusted R^2	0.397	0.396	0.344

Application I: Gradient and Placebo

Panel B: Marginal effect of physical similarity by level of shared dependence

1 SD below average	−0.0042 (0.0027)	0.0051 (0.0033)
Average dependence	0.0012 (0.0025)	0.0034 (0.0027)
1 SD above average	0.0066** (0.0032)	0.0016 (0.0031)
2 SD above average	0.0119*** (0.0044)	−0.0001 (0.0042)

Panel C: Difference between the nature and placebo gradients

	Estimate	Std. error	p-value
Nature gradient	0.0054	(0.0016)	0.0010
Governance gradient (placebo)	−0.0017	(0.0017)	0.3002
Difference (nature – governance)	0.0071	(0.0021)	0.0006
Stacked observations		643,360	

Application II: Baseline

<i>Panel A: Baseline estimates</i>				
	Applicant-added citation		Examiner-added (placebo)	
	All pairs (1)	First links (2)	All pairs (3)	First links (4)
Physical similarity	0.1634*** (0.0333)	0.0184*** (0.0046)	0.0865*** (0.0280)	0.0045 (0.0036)
Technological proximity	0.9760*** (0.1005)	0.0865*** (0.0111)	1.1521*** (0.1318)	0.1082*** (0.0168)
Distance between facility portfolios (log km)	-0.0125 (0.0312)	-0.0020 (0.0061)	-0.0315 (0.0315)	-0.0051 (0.0056)
Same state	-0.3413*** (0.1082)	-0.0133 (0.0198)	-0.2203* (0.1221)	-0.0124 (0.0191)
Knowledge stocks and patent counts	Yes	Yes	Yes	Yes
Facility counts	Yes	Yes	Yes	Yes
Firm-difference controls	Yes	Yes	Yes	Yes
Industry $\times$ year FE (both firms)	Yes	Yes	Yes	Yes
Observations	607,027	596,641	607,027	596,641
Mean of dependent variable (%)	0.6420	0.0679	0.6102	0.0582
Citation events	3,905	406	3,712	348
Firms	1,268	1,268	1,268	1,268
Adjusted R^2	0.056	0.010	0.064	0.009

Application II: Robustness

<i>Panel B: Robustness, coefficient on physical similarity</i>				
Headquarters > 1,000 km apart	0.1956*** (0.0400)	0.0238*** (0.0059)	0.1168*** (0.0360)	0.0074 (0.0050)
Different state as well	0.1792*** (0.0362)	0.0209*** (0.0048)	0.0934*** (0.0288)	0.0052 (0.0038)
No facility within 25 km	0.0430*** (0.0160)	0.0050 (0.0033)	−0.0024 (0.0132)	−0.0014 (0.0025)
Firm × year fixed effects	0.1003*** (0.0351)	0.0112** (0.0055)	0.0478 (0.0328)	0.0009 (0.0051)

Significant in seven of eight years. Leave-one-year-out t : 4.57 to 5.28. 200 permutations: largest t 2.81, against 4.90 in the data.

Known Gaps

Gap	Why
No measurement-error estimate	no second reading to instrument with
Coverage needs facility coordinates	2,482 firms, tilted to multi-establishment
United States only, 2017 to 2024	the embedding layers start in 2017
Environmental scores lose precision	cross-industry, toxic emissions only